

Question 2 (12 marks). Use a SEPARATE Writing Booklet.

Marks

- | | | |
|-----|---|---|
| (a) | Evaluate $ -7 - -9 $ | 2 |
| (b) | Given $f(x) = x - \frac{1}{x}$ evaluate $f(\frac{1}{2})$ | 1 |
| (c) | Find the area of the triangle formed by the line $3x + 8y - 4 = 0$ and the x and y axes. | 2 |
| (d) | Solve for x : $ x - 3 > 2$ | 2 |
| (e) | Write down the equation of the circle with centre at the origin and passing through the point $(5,7)$. | 2 |
| (f) | Find the equation of the line passing through the point $(3,0)$ and the point of intersection of the lines $x + y + 1 = 0$ and $2x - y - 4 = 0$. | 3 |

Question 3 (12 marks). Use a SEPARATE Writing Booklet.

- | | | |
|-----|--|---|
| (a) | Given that $\sin \theta = \frac{3}{\sqrt{15}}$ and $\cos \theta < 0$, find the exact value of $\tan \theta$. | 2 |
| (b) | Simplify $\frac{\sin(90^\circ - \theta)}{\sin(180^\circ - \theta)}$. | 2 |
| (c) | Solve for θ : $\sin \theta = -0.5$ for $0 \leq \theta \leq 360^\circ$ | 2 |
| (d) | In an isosceles triangle the equal sides measure 15cm and the equal angles measure 37° . Find the length of the remaining side, correct to 2 significant figures. | 3 |
| (e) | Draw, on separate axes, a possible graph of the quadratic function of the form $y = ax^2 + bx + c$. | |
| | (i) if $a < 0$ and $\Delta > 0$ | 1 |
| | (ii) if $a > 0$ and $\Delta < 0$ | 1 |
| | (iii) if $a > 0$ and $\Delta = 0$ | 1 |

Question 4 (12 marks) Use a SEPARATE Writing Booklet.

Marks

On separate number planes, draw neat sketches, showing important features, of the following:

- | | | |
|-----|--|---|
| (a) | $f(x) = 2 - x^2$ | 2 |
| (b) | $f(x) = (x - 1)^2$ | 2 |
| (c) | $x^2 + y^2 = 5$ | 2 |
| (d) | $y = \frac{1}{x + 2}$ | 2 |
| (e) | $f(x) = \begin{cases} x^2, & \text{for } x \leq 0 \\ x + 1, & \text{for } x > 0 \end{cases}$ | 2 |
| (f) | $f(x) = \sqrt{4 - x}$ | 2 |

Question 5 (12 marks) Use a SEPARATE Writing Booklet.

(a) Differentiate the following functions:

- | | | |
|-------|--|---|
| (i) | $y = x^2 + \frac{1}{3x^3}$ | 2 |
| (ii) | $y = \frac{x}{2x - 1}$ | 2 |
| (iii) | $y = (x^2 - x)^4$ | 2 |
| (iv) | $y = x^3(x - 1)^5$, leaving your answer in fully factorised form. | 3 |

(b) What can be said about the function $y = f(x)$ if:

- | | | |
|-------|-------------------------------------|---|
| (i) | $f'(x) = 0$ for all values of x . | 1 |
| (ii) | $f'(x)$ is a linear function. | 1 |
| (iii) | $f(a) = -f(-a)$ | 1 |

Question 6 (12 marks). Use a SEPARATE Writing Booklet.

- | | | |
|-------|---|---|
| (a) | Given the points $P(2,-2)$ and $Q(-4,6)$, | |
| (i) | find the length of the interval PQ | 1 |
| (ii) | show that the midpoint of PQ is $(-1,2)$. | 1 |
| (iii) | find the equation of the straight line joining the point $(4,-3)$ to the midpoint of PQ . | 2 |
| (b) | Find the equation of the straight line that is perpendicular to $y = 5x + 2$ and passes through $(1,5)$. | 2 |
| (c) | Find distance from the point $(1,-2)$ to the line $y = x + 6$ | 3 |
| (d) | Shade the region on the number plane for which $y \geq x^2 + 4x$ and $y \leq x + 1$ hold simultaneously. | 3 |

Question 7 (12 marks) Use a SEPARATE Writing Booklet.

- | | | |
|-------|--|---|
| (a) | Given the quadratic function $y = x^2 + x + 1$ whose roots are α and β , write down the value of: | |
| (i) | $\alpha + \beta$ | 1 |
| (ii) | $\alpha\beta$ | 1 |
| (iii) | $\frac{1}{\alpha} + \frac{1}{\beta}$ | 2 |
| (iv) | $\alpha^2 + \beta^2$ | 2 |
| (b) | Given a function $y = x^2 + x$, | |
| (i) | find the gradient of the tangent to the curve at the point $A (-1,0)$. | 2 |
| (ii) | show that the equation of the normal at A is $x - y + 1 = 0$. | 2 |
| (iii) | find the coordinates of the point B where the normal meets $y = x^2 + x$ again. | 2 |

Question 8 (12 marks) Use a SEPARATE Writing Booklet.

Marks

(a) Factorise fully, the following:

(i) $x^4 + 125x$

2

(ii) $40x^2 + 11x - 2$

2

(iii) $100 - (40 - t)^2$

2

(b) State the largest possible domain for the function $f(x) = \frac{3}{x^2 - 4}$

2

(c) What is the minimum value of $f(x) = x^2 - 10x + 21$?

2

(d) If $f(x) = x^2 - 2x + 4$ find the value(s) of x for which $f(x) = 7$

2

Question 9 (12 marks). Use a SEPARATE Writing Booklet.

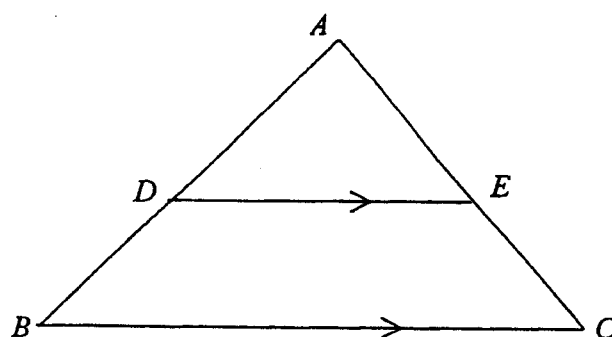
(a) Solve $(3^x - 1)(3^x - 9) = 0$

2

(b) Explain why there is no solution to $2 - \cos x = 0$

1

(c)



In the diagram, ABC is an isosceles triangle where $AB = AC$.
 DE is parallel to BC .

(i) Show that $\triangle ADE$ is an isosceles triangle.

3

(ii) Show that $DB = EC$.

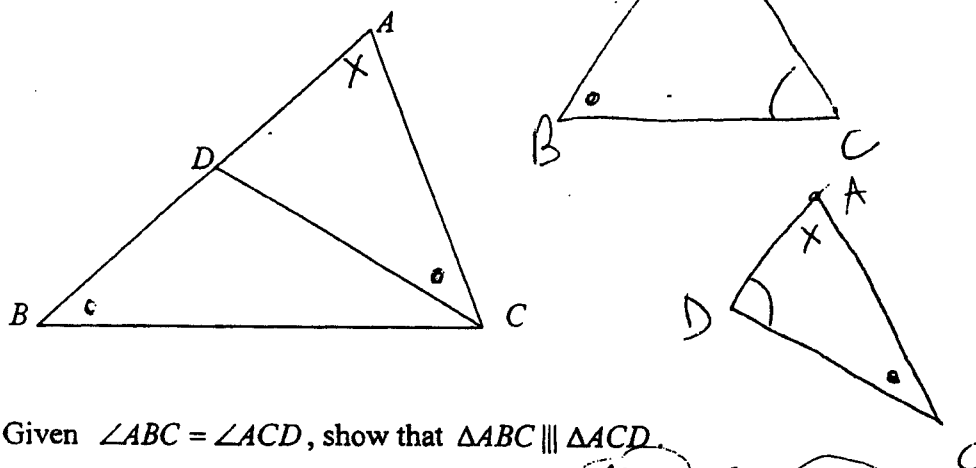
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Question 9(d) is on the next page.

Question 9 (continued)

(d)

Mark



(i) Given $\angle ABC = \angle ACD$, show that $\triangle ABC \parallel \triangle ACD$.

(ii) Hence show that $AC^2 = AB \times AD$.

(iii) Find the length of AB if $AC=6\text{cm}$, $AD=4\text{cm}$.

$$\frac{AB}{AC} = \frac{BC}{CD} = \frac{AC}{AD}$$

Question 10 (12 marks) Use a SEPARATE Writing Booklet.

(a) Show that $2 \sin x - \cos^2 x + 2 = (\sin x + 1)^2$

(b) (i) Write down an expression for the discriminant of the quadratic equation $x^2 + 5x + a^2 = 0$.

(ii) Hence, find the value(s) of a for which the function $f(x) = x^2 + 5x + a^2$ is positive definite.

(c) A ship sails from port A , 30km due East to port B . It then sails a further 18km in the direction 155°T to port C .

(i) Draw a neat sketch of the above information.

(ii) Show that $\angle ABC = 115^\circ$.

(iii) Find the distance from port A to port C , correct to the nearest metre.

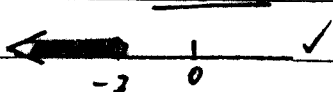
(iv) Find the bearing of port C from port A , correct to the nearest degree.

End of Paper.

YEAR 11 MATHEMATICS YEARLY EXAM 2002

1.(a) 6.89

(b) $1-3x > 10$
 $-3x > 9$

$x < -3$ ✓


(c) $\sin 300^\circ = -\sin 60^\circ$
 $= -\frac{\sqrt{3}}{2}$

(d) $4 \cdot 1 \times 10^{-3}$

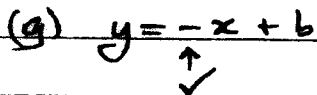
(e) $\frac{2x}{3} - \frac{x+1}{4} = 1$

$8x - 3x - 3 = 12$ ✓

$5x = 15$

$x = 3$ ✓

(f) $\frac{1}{1-\sqrt{2}} \times \frac{1+\sqrt{2}}{1+\sqrt{2}} = \frac{1+\sqrt{2}}{1-2}$
 $= -1-\sqrt{2}$

(g) $y = -x + 6$


2(a) $|-7| - |-9| = 7-9$
 $= -2$

(b) $\frac{1}{2} - \frac{1}{\frac{1}{2}} = -1\frac{1}{2}$

(c) $x=0$ $y=\frac{1}{2}$
 $y=0$ $x=\frac{4}{3}$

$A = \frac{1}{2} \cdot \frac{4}{3}$ ✓
 $= \frac{2}{3}$ (sq units) ✓

(d) $|x-3| > 2$

$x-3 < -2$, $x-3 > 2$

$x < 1$ ✓ $x > 5$ ✓

(e) $z = \sqrt{8^2 + 7^2}$
 $= 11$

$x^2 + y^2 = 4$

f) $x+y+1+k(2x-y-4)=0$

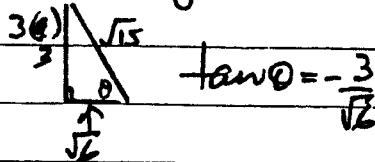
$3+1+k(6-4)=0$

$k=2$ ✓

$x+y+1-2(2x-y-4)=0$ ✓

$x+y+1-4x+2y+8=0$

$-3x+3y+9=0$ ✓

3(a)  $\tan \theta = -\frac{3}{\sqrt{13}}$

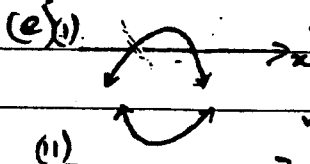
(b) $\frac{\sin(90-\theta)}{\sin(80-\theta)} = \frac{\cos \theta}{\sin \theta}$ ✓
 $= \cot \theta$ ✓

(c) $\sin \theta = -0.5$

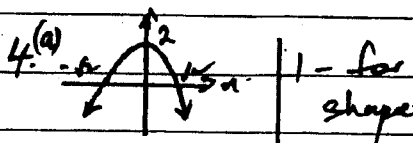
" θ " = 30° / 40° ✓

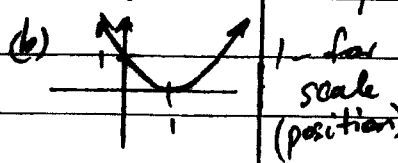
$\theta = 210^\circ, 330^\circ$ ✓

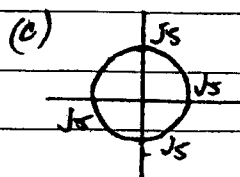
(d) $L = 15 + 15 - 2 \cdot 15 \cdot \cos 37^\circ$
 $= 450 - 23.95 \dots$
 $= 20.64 \dots$
 $= 21 \text{ cm.}$ ✓ (24)

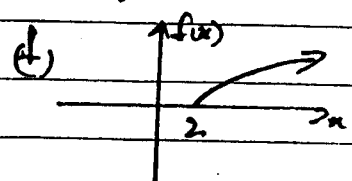
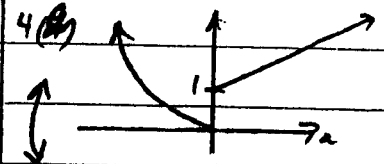
(e)(i) 

(ii) 

4(a)  1 - for shape

(b)  1 - for scale (position)

(c) 



5(a) $y = x^2 + \frac{1}{3}x^{-3}$
 $\frac{dy}{dx} = 2x - x^{-4}$ ✓ ✓

(ii) $y = \frac{x}{2x-1}$

$\frac{dy}{dx} = \frac{(2x-1) \cdot 1 - x \cdot 2}{(2x-1)^2}$ ✓
 $= \frac{-1}{(2x-1)^2}$ ✓

(iii) $y = (x^2-x)^4$
 $\frac{dy}{dx} = 4(x^2-x)^3 \cdot (2x-1)$ ✓ ✓

(iv) $y = x^3(x-1)^5$
 $\frac{dy}{dx} = x^3 \cdot 5(x-1)^4 \cdot 1 + (x-1)^5 \cdot 3x^2$ ✓ ✓

$= x^2(x-1)^4(5x+3x-3)$

$= x^2(x-1)^4(8x-3)$ ✓

(b) (i) $f(x)$ is linear (or example)
 or $f(x)$ is a horizontal line

(ii) $f(x)$ is a str. line (or eq.)

(iii) $f(x)$ is odd.

$$6(a)(i) PQ = \sqrt{(6+2)^2 + (-4-2)^2}$$

$$= 10. \quad \checkmark$$

$$(ii) M \left(\frac{2+4}{2}, \frac{-2+6}{2} \right) \checkmark$$

$$(-1, 2).$$

$$(iii) y-3 = \frac{2+3}{-184} (x-4) \checkmark$$

$$y+3 = -x+4$$

$$y = -x+1 \quad (\text{or variation})$$

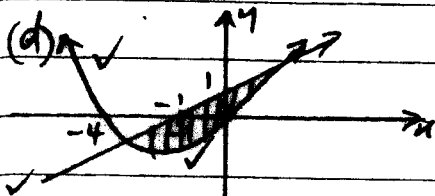
$$(b) y-5 = -\frac{1}{3}(x-1)$$

$$y = -\frac{1}{3}x + \frac{5}{3}$$

$$\text{or } x+5y-26=0.$$

$$(c) d = \frac{|1.1 + -1.2 - 6|}{\sqrt{(-1)^2 + 1^2}} \checkmark \checkmark$$

$$= \frac{3}{\sqrt{2}} \text{ or } \frac{3\sqrt{2}}{2}$$



$$7(a) \alpha + \beta = -1 \quad \checkmark$$

$$(ii) \alpha\beta = 1 \quad \checkmark$$

$$(iii) \beta + \alpha = -1 = -1 \quad \checkmark$$

$$(iv) (\alpha + \beta) - 2\alpha\beta = (-1) - 2 = -3 \quad \checkmark$$

$$(b) y = x^2 + x$$

$$\frac{dy}{dx} = 2x + 1 \quad \checkmark$$

$$(i) m = -1 \quad \checkmark$$

$$(ii) y = -x + b \quad \checkmark$$

different question? $0 = -1 + b$

$$y = -x + 1$$

$$x + y + 1 = 0 \quad \checkmark$$

$$(iii) x^2 + x = x + 1$$

$$x^2 = 1$$

$$x = \pm 1$$

$$\text{meets at } (1, 2) \quad \checkmark$$

$$(1, 2) \quad \checkmark$$

$$8(a)(i) x^2 + 15x = x(x^2 + 15)$$

$$= x(x+5)(x-5+25) \quad \checkmark \checkmark$$

$$(ii) 40x^2 + 11x - 2 \quad \checkmark$$

$$= 40x^2 + 16x - 5x - 2 \quad \checkmark$$

$$= 8x(5x+2) - (5x+2) \quad \checkmark$$

$$= (8x-1)(5x+2) \quad \checkmark \checkmark$$

$$(iii) 100 - (40-t)^2$$

$$(10-40+t)(10+40-t)$$

$$(t-30)(50-t) \quad \checkmark \checkmark$$

$$(b) x^2 - 4 \neq 0 \quad \checkmark$$

$$x \neq \pm 2, \quad \checkmark$$

$$A: \text{all real } x, \text{ except } x = \pm 2.$$

$$(c) x = \frac{10}{2} = 5.$$

$$f(5) = 25 - 50 + 21$$

$$= -4 \leftarrow \text{min value.}$$

$$(d) x^2 - 2x + 4 = 7$$

$$x^2 - 2x - 3 = 0$$

$$(x-3)(x+1) = 0$$

$$x = 3, -1$$

$$9(a) (3^x - 1)(3^x - 9) = 0$$

$$3^x = 1 \quad 3^x = 9$$

$$\text{then } x = 0 \quad \checkmark \quad x = 2 \quad \checkmark$$

$$(b) -1 \leq \cos x \leq 1. \quad \checkmark$$

$$(c) (i) \angle ABC = \angle ACB \quad (\text{base } \angle s) \quad \checkmark$$

$$\angle ADE = \angle ABC \quad (\text{corr } \angle s) \quad \checkmark$$

$$\angle AED = \angle ACB \quad (\text{corr } \angle s) \quad \checkmark$$

$$\therefore \angle ADE = \angle AED \quad (\text{from above}) \quad \checkmark$$

$$\text{i.e. } \triangle ADE \text{ is isos. (base } \angle s \text{ equal)} \quad \checkmark$$

$$(ii) \angle ADB = \angle AEC \quad (\text{vert } \angle s) \quad \checkmark$$

$$AD = AE \quad (\text{isos } \triangle ADE) \quad \checkmark$$

$$DB = EC \quad (\text{by subtraction}) \quad \checkmark$$

if using III they need to write correct ratios + then say $AB = AC$ then $DB = EC$ from ratios.

OR show DE || BC as isos. triangles

2

1

can use III but need to show sides equal first

if using III they need to write correct ratios + then say $AB = AC$ then $DB = EC$ from ratios.

9. (a) In Δ 's ABC & ACD .

(i) $\angle BAC = \angle CAD$ (common) ✓

$\angle ABC = \angle ACD$ (given)

$\therefore \angle ACB = \angle ADC$ (angle sum of Δ) ✓

$\Delta ABC \parallel \Delta ACD$ (equiangular).

(ii) $\frac{AC}{AD} = \frac{AB}{AC}$ (corresponding side of similar Δ 's). ✓

$AC^2 = AB \cdot AD$.

(iii) $6^2 = AB \cdot 4$

$AB = 9$ ✓

10(a) $LHS = 2\sin x - \cos^2 x + 2$

$= 2\sin x - (1 - \sin^2 x) + 2$

$= \sin^2 x + 2\sin x + 1$ ✓

$= (\sin x + 1)^2$

(b) (i) $A = 25 - 4a^2$

(ii) $25 - 4a^2 < 0, a > 0$.

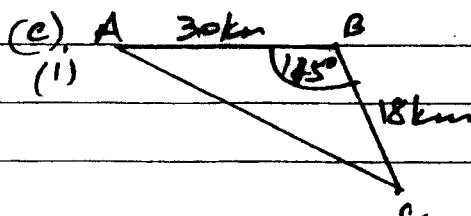
$4a^2 > 25$ ✓

$a^2 > 25$

$a < -5, a > 5$ but $a > 0$.

$\therefore a > 5$ ✓

$a > 5/2$.



(ii) $\angle ABC = 270^\circ - 155^\circ$ ✓
 $= 115^\circ$

(iii) $AC^2 = 30^2 + 18^2 - 2 \cdot 30 \cdot 18 \cos 115^\circ$ ✓

$= 1680.42 \dots$

$AC = 50m$ ✓

(iv) $\frac{\sin A}{18} = \frac{\sin 115^\circ}{50}$ ✓

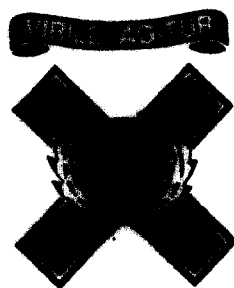
$\sin A = \frac{18 \sin 115^\circ}{50}$

$= 0.326 \dots$ ✓

$A = 19^\circ$ ✓

EH

EDB



KNOX GRAMMAR SCHOOL
MATHEMATICS DEPARTMENT

2002
YEARLY EXAMINATION

Mathematics

Year 11

Total marks (120)



• General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using blue or black pen
- Board-approved calculators may be used
- All necessary working should be shown in every question
- Attempt Questions 1–10
- All questions are of equal value
- Use a separate Writing Booklet for each question
- An answer booklet must be handed in for each question.
- If a question is not attempted please write N/A on the front of the answer booklet

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Total marks (120)

Attempt questions 1 – 10

All questions are of equal value

Answer each question in a SEPARATE Writing Booklet. Extra Writing Booklets are available.

	Marks
Question 1 (12 marks) Use a SEPARATE Writing Booklet.	
(a) Find the value of $(2^{0.7} + 1)^2$ correct to two decimal places.	1
(b) Solve $1 - 3x > 10$ and graph the solution on the number line.	2
(c) Give the <u>exact</u> value of $\sin 300^\circ$	2
(d) Express the answer to 0.0034×1.2 in scientific notation, correct to 2 significant figures.	1
(e) Solve for x : $\frac{2x}{3} - \frac{x+1}{4} = 1$	2
(f) Find integers a and b such that $\frac{1}{1-\sqrt{2}} = a - \sqrt{b}$.	3
(g) Write down the equation of the straight line inclined at 135° to the positive x axis.	1